

# Large Margin Classifiers

## ECMM422

Fabrizio Costa

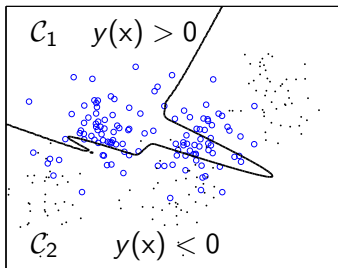
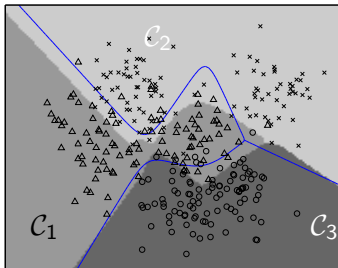
Winter 2026



# Discriminants

Define a function  $y_k(x)$  so that if  $y_k(x) > y_j(x)$  for all  $k \neq j$  then  $x$  is classified as class  $C_k$ .

*Blue line is Bayes optimal decision boundary*



With only two classes, define

$$y(x) = y_1(x) - y_2(x)$$

Decision boundary is  $y(x) = 0$

# Linear discriminants

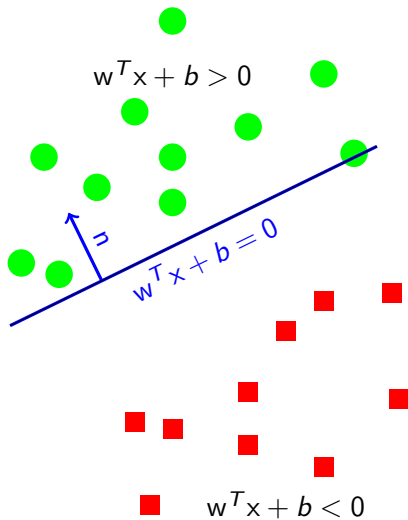
For two classes:

$$y(x) = w^T x + b$$

$$= \sum_{i=1}^D w_i x_i + b$$

for some *weights*  $w$  and an offset or bias  $b$

- ▶  $w^T x + b$  defines a (hyper-) plane
- ▶ Unit vector normal to hyper-plane is  $n = w / \|w\|$



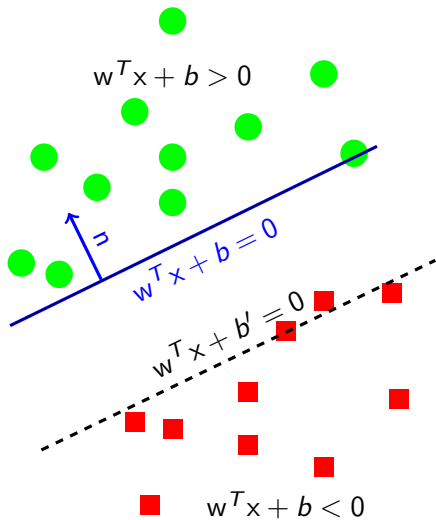
# Linear discriminants

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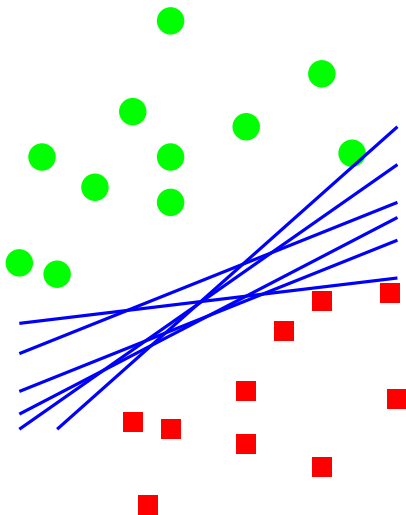
for some *weights*  $w$  and an offset or bias  $b$

- ▶  $w^T x + b$  defines a (hyper-) plane
- ▶ Unit vector normal to hyper-plane is  $n = w/\|w\|$
- ▶ Bias  $b$  controls distance of decision boundary from origin: perpendicular distance to origin is  $b/\|w\|$



# Which discriminant?

- ▶ How to decide on which discriminant to use?
- ▶ Generally an infinite range of possibilities
- ▶ One possibility would be to minimise the *cross-entropy* between targets and posterior probabilities  $p(t | x)$ . Requires a model to produce posterior probabilities.



Linear discriminant decision boundary between  $\mathcal{C}_j$  and  $\mathcal{C}_k$  is determined by locus

$$y_1(x) = y_2(x)$$

If  $g(\cdot)$  is monotone

$$g(y_1(x)) = g(y_2(x))$$

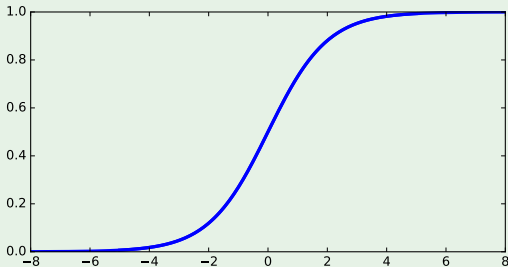
so decision boundary is unaltered with discriminant:  $g(y(x))$

$$a = \mathbf{w}^T \mathbf{x} + b$$

$$y = g(\mathbf{w}^T \mathbf{x} + b)$$

Activation/link function  $g(a)$  is sigmoidal:

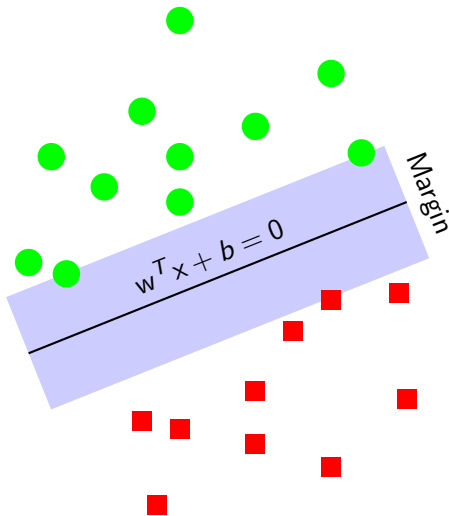
$$g(a) = \frac{1}{1 + e^{-a}}$$



- ▶ Ensures  $0 < y(\mathbf{x}) < 1$  so output may be interpreted as a probability of class membership
- ▶ Train by minimising the cross entropy with respect to  $\mathbf{w}$  and  $b$ .

# Large margin linear classifier

- ▶ Choose discriminant to maximise the *margin*
- ▶ Good theoretical properties and empirical performance



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- ▶ Choose discriminant to maximise the *margin*
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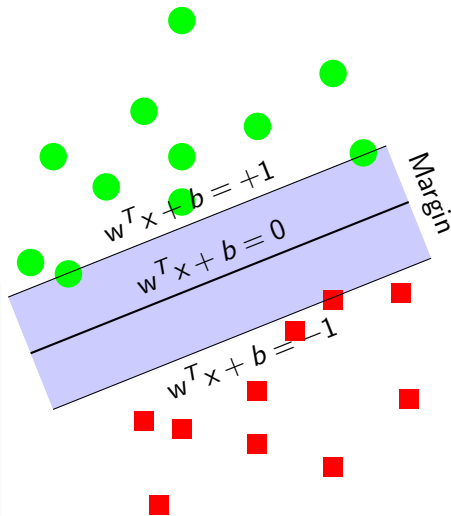
Data:  $\{(x_n, t_n)\}_{n=1}^N$  with  $t_n \in \{-1, 1\}$

Maximise margin by

$$\max_{w, b} \frac{2}{\|w\|} \quad \text{distance between hyperplanes}$$

$$\text{s.t. } w^T x_n + b \geq +1, \quad t_n = +1$$

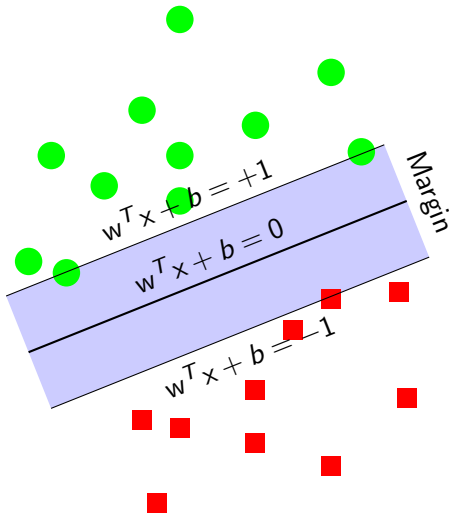
$$w^T x_n + b \leq -1, \quad t_n = -1$$



# Large margin linear classifier

Equivalent formulations:

$$\begin{aligned} \max_{w,b} \quad & \frac{2}{\|w\|} \\ \text{s.t.} \quad & t_n(w^T x_n + b) \geq 1 \end{aligned}$$

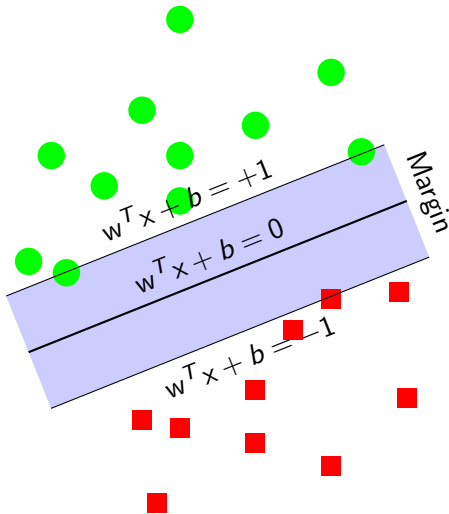


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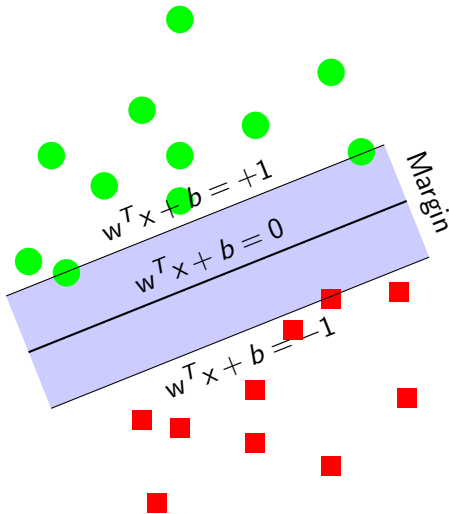


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- ▶ Quadratic (convex) programme with linear constraints
- ▶ Lagrangian duality to maximise the *Lagrangian dual function*

## Lagrangian dual function

$$g(\alpha) = \min_{w,b} L(w, b, \alpha) = \min_{w,b} \left[ \frac{1}{2} \|w\|^2 - \sum_{n=1}^N \alpha_n (t_n (w^T x_n + b) - 1) \right]$$

$$\frac{\partial L(w, b, \alpha)}{\partial w} = 0 \quad \Rightarrow \quad w = \sum_{n=1}^N \alpha_n t_n x_n$$

$$\frac{\partial L(w, b, \alpha)}{\partial b} = 0 \quad \Rightarrow \quad \sum_{n=1}^N \alpha_n t_n = 0$$

# Lagrangian dual

---

$$\mathbf{w}^* = \sum_{n=1}^N \alpha_n t_n \mathbf{x}_n \quad \text{and} \quad \sum_{n=1}^N \alpha_n t_n = 0$$

$$g(\boldsymbol{\alpha}) = \frac{1}{2} \|\mathbf{w}^*\|^2 - \sum_{n=1}^N \alpha_n (t_n (\mathbf{w}^{*T} \mathbf{x}_n + b) - 1)$$

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$$g(\boldsymbol{\alpha}) = \frac{1}{2} \|\mathbf{w}^*\|^2 - \sum_{n=1}^N \alpha_n t_n \left( \left[ \sum_{m=1}^N \alpha_m t_m \mathbf{x}_m \right]^T \mathbf{x}_n \right) + \sum_{n=1}^N \alpha_n$$

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$$g(\alpha) = \sum_{n=1}^N \alpha_n - \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N t_n t_m \alpha_n \alpha_m \mathbf{x}_n^T \mathbf{x}_m$$

# Lagrangian dual problem

$$\max_{\alpha} \sum_n \alpha_n - \frac{1}{2} \sum_{m=1}^N \sum_{n=1}^N t_n t_m \alpha_n \alpha_m \langle \mathbf{x}_n, \mathbf{x}_m \rangle$$

such that  $\alpha_n \geq 0, \quad \sum_n \alpha_n t_n = 0$

- ▶  $\langle \mathbf{x}_n, \mathbf{x}_m \rangle$  is the inner product:  $\mathbf{x}_n^T \mathbf{x}_m$
- ▶ The quadratic programme with linear constraints can be numerically solved efficiently

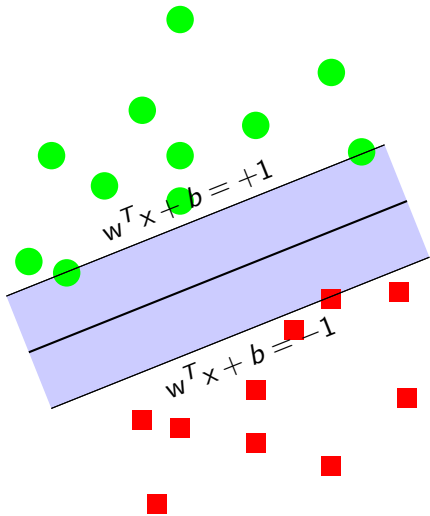
# Support vectors

## KKT conditions

$$\alpha_n(t_n(w^T x_n + b) - 1) = 0$$

$$w = \sum_n \alpha_n t_n x_n$$

$$\sum_n \alpha_n t_n = 0$$



# Support vectors

## KKT conditions

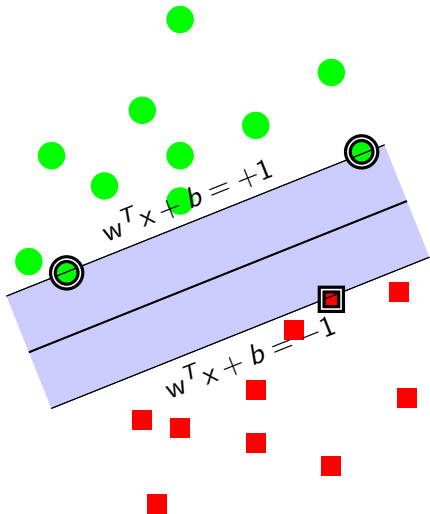
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## Support vectors

- ▶ Non-zero  $\alpha_n$  occur only at the edges of the margin
- ▶  $x_n$  for which  $\alpha_n > 0$  are called *support vectors*



# Support vectors

## KKT conditions

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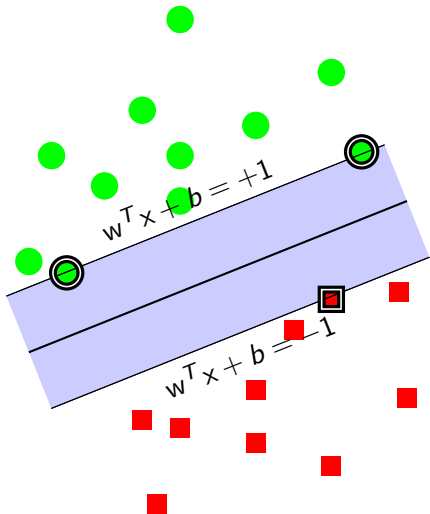
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## Support vectors

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- ▶  $x_n$  for which  $\alpha_n > 0$  are called *support vectors*

- ▶ Compute  $w$  from support vectors and  $b$  from  $t_n(w^T x_n + b) - 1 = 0$



# Computing the discriminant function

Classify  $x$  according to the sign of discriminant function:

$$y(x) = w^T x + b$$

$$w = \sum_n \alpha_n t_n x_n$$

Discriminant function

$$\begin{aligned} y(x) &= \sum_n \alpha_n t_n x_n^T x + b \\ &= \sum_{n \in SV} \alpha_n t_n \langle x_n, x \rangle + b \end{aligned}$$

- ▶ Sum only runs over the support vectors
- ▶ No need to form the, possibly very high dimensional,  $w$  explicitly

# Soft margin SVM

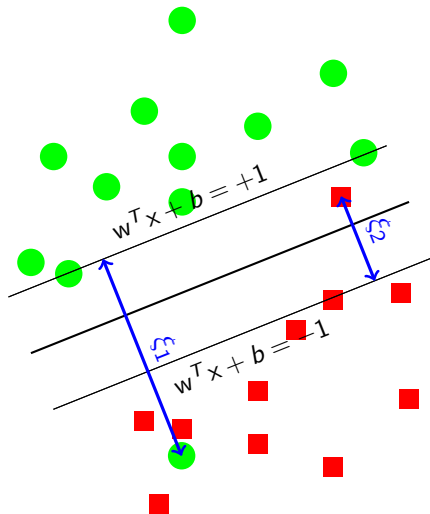
- ▶ If data is not *linearly separable* add *slack variables*  $\xi_n$  associated to each  $x_n$

## Soft margin SVM

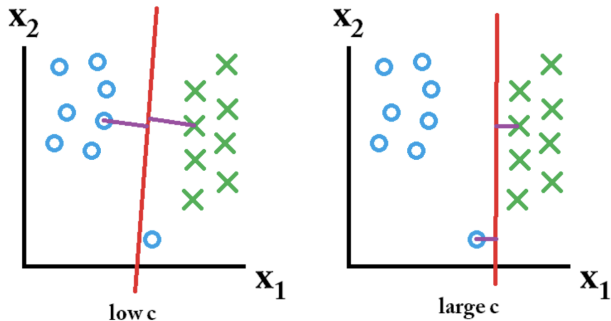
$$\min_{w,b} \frac{1}{2} \|w\|^2 + C \sum_n \xi_n$$

$$\text{s.t. } t_n(w^T x_n + b) \geq 1 - \xi_n$$
$$\xi_n \geq 0 \text{ for any } n$$

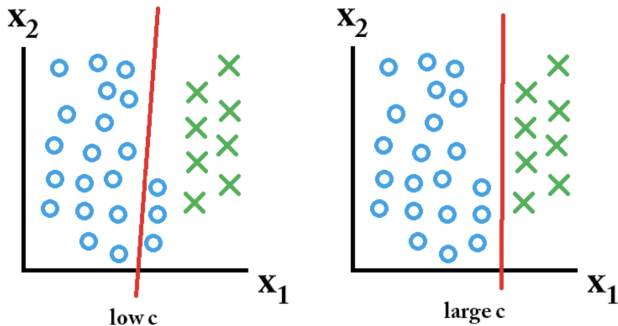
- ▶  $C$  is a *regularisation parameter* trading off quality of fit to data with width of margin



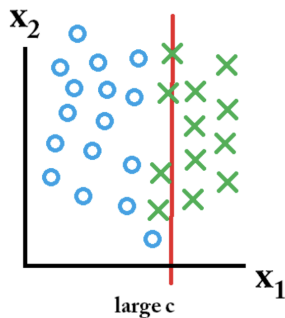
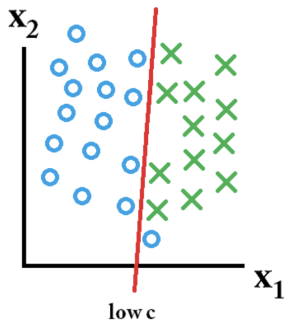
# Soft margin influence of $C$



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## Lagrangian dual function

$$\max_{\alpha} \quad \sum_n \alpha_n - \frac{1}{2} \sum_{n,m} t_n t_m \alpha_n \alpha_m \langle \mathbf{x}_n, \mathbf{x}_m \rangle$$

such that  $0 \leq \alpha_n \leq C, \quad \sum_n \alpha_n t_n = 0$

- ▶  $\langle \mathbf{x}_n, \mathbf{x}_m \rangle$  is the inner product:  $\mathbf{x}_n^T \mathbf{x}_m$
- ▶ The constraint  $\alpha_n < C$  is necessary to prevent  $C - \alpha_n$  from becoming negative and allowing to drive the cost to  $-\infty$
- ▶ The quadratic programme with linear constraints can be numerically solved efficiently

## Lagrangian dual function

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Only inner products are needed to form objective function  
and to solve resulting optimisation problem

Note that in this case the KKT conditions will give us

$$\alpha_n = 0 \Rightarrow t_n(w^T x_n + b) \geq 1$$

$$\alpha_n = C \Rightarrow t_n(w^T x_n + b) \leq 1$$

$$0 < \alpha_n < C \Rightarrow t_n(w^T x_n + b) = 1$$

i.e. the set of support vectors now includes all data points on the margin boundary as well as those on the wrong side of the margin boundary.